

# Notes: Subrahmanyam, Tang, and Wang (RFS, 2014)

## Does the Tail Wag the Dog?: The Effect of Credit Default Swaps on Credit Risk

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# 1 Big picture

- **Question.** Does the introduction of credit default swaps (CDS) on a firm's debt increase the firm's credit risk?
- **Setting.** The paper studies North American firms from 1997 to 2009 using actual CDS transaction data merged with ratings, Compustat–CRSP accounting and market data, bankruptcy filings, and syndicated-loan information.
- **Core challenge.** CDS trading is not randomly assigned. Large, visible, relatively high-quality firms are more likely to attract CDS trading in the first place, and market participants may also start CDS trading precisely when they anticipate future credit deterioration.
- **Main conceptual contribution.** The paper studies the *real effect* of CDS on the underlying borrower rather than only the pricing of the derivative itself. Its organizing mechanism is the *empty creditor* problem: once creditors hedge default risk with CDS, their incentives in renegotiation can become tougher and can tilt distressed firms toward bankruptcy.
- **Main hypotheses.**
  - **Baseline:** credit risk rises after CDS inception.
  - **Conditional on distress:** once a firm is distressed, CDS exposure makes bankruptcy more likely.
  - **No restructuring clause:** the effect should be stronger when CDS only pay in bankruptcy or failure-to-pay states and not in restructuring.
  - **Creditor coordination:** CDS availability broadens the lender base, which worsens coordination in workouts.
- **Main empirical design.** The baseline specification is a monthly discrete-time hazard/logit model for rating downgrades and bankruptcy, with *CDS Active* equal to one after CDS trading begins. The paper then addresses endogeneity with (i) a CDS-selection probit, (ii) propensity-score matching, and (iii) full-information maximum likelihood with instruments.
- **Main baseline result.** After CDS trading begins, firms become significantly more likely to be downgraded and to go bankrupt. In the bankruptcy regressions, the marginal effect is economically large relative to the very low unconditional bankruptcy probability.
- **Main mechanism results.** Bankruptcy risk rises with the amount of CDS outstanding relative to debt, rises especially for distressed firms, rises more when the contract is of the “no restructuring” type, and rises with the number of bank creditors that appears after CDS introduction.
- **Main credibility result.** The positive CDS effect survives observable selection adjustment, matched-sample estimation, and joint estimation that treats CDS introduction as endogenous.
- **Economic takeaway.** CDS are not purely side bets. By changing creditor incentives and creditor structure, they can feed back into the default risk of the very firms on which they are written—the “tail” can wag the “dog.”

## 2 Data and measurement (key objects)

## 2.1 Why this setting is useful

- The paper studies the broad inception phase of the corporate CDS market, beginning in 1997, so first trading dates are not contaminated by starting the sample after the market was already mature.
- The data are based on *actual transactions*, not only dealer quotes. That matters because the paper needs accurate timing for CDS inception and detailed information on contract terms.
- The main outcome, bankruptcy filing, is a clear real event. That makes the paper sharper than papers that only study spreads or ratings.

**Interpretation.** This is a setting where the authors can link a derivative market directly to hard credit outcomes of the underlying firms.

## 2.2 CDS transaction data and sample construction

The CDS data come from:

- **CreditTrade**, covering June 1997 to March 2006,
- **GFI Group**, covering January 2002 to April 2009,
- and cross-checks with **Markit** quotes.

Main sample facts:

- 901 North American firms experience CDS inception during 1997–2009.
- CDS introductions are concentrated in 2000–2003.
- The paper constructs a comprehensive bankruptcy sample with 1,628 bankruptcy filings from 1997 to 2009.
- In the final regression sample, there are 940 distinct bankrupt firms, of which 42 are CDS firms.

**Interpretation.** CDS firms are a selected subset of relatively large, visible corporate borrowers, so the paper cannot rely on raw comparisons alone.

## 2.3 Credit-risk outcomes

The paper uses two main measures of credit risk:

- **Credit rating downgrades**, based on S&P long-term issuer ratings from Compustat and FISD.
- **Bankruptcy filings**, assembled from BankruptcyData.com, the UCLA-LoPucki database, the Altman-NYU list, FISD, and Moody's recovery reports.

In the preliminary rating event study:

- 92% of firms are rated at CDS inception.
- 54% keep the same rating from  $t - 1$  to  $t + 2$ .

- 37% are downgraded, and only 9% are upgraded.
- In matched diff-in-diff style analysis, downgrades for CDS firms are about 0.23 notches larger than for matched non-CDS firms, and downgrading frequency is 2.6 percentage points higher.

**Interpretation.** Ratings already suggest that the credit quality of CDS-referenced firms worsens after CDS begin trading.

## 2.4 Main CDS variables

The paper uses several CDS measures.

### 1. Treatment indicator.

- **CDS Active:** equals one after a firm's CDS trading begins and zero before.
- **CDS Firm:** equals one for firms that ever have CDS in the sample.

### 2. Intensity measure.

- **CDS Notional Outstanding/Total Debt:** notional CDS outstanding divided by the firm's book debt. This scales CDS positions by the size of creditors' exposure.

### 3. Contract-type measure.

- **No Restructuring CDS:** the fraction of active CDS contracts on the firm that exclude restructuring as a credit event.

**Interpretation.**

- *CDS Active* asks whether the regime change matters.
- CDS notional outstanding asks whether more CDS exposure matters more.
- *No Restructuring CDS* isolates the empty-creditor channel more directly.

## 2.5 Fundamental controls in the baseline credit-risk model

The baseline bankruptcy and downgrade regressions follow Bharath and Shumway (2008) and include:

- $\ln(E)$ : log market value of equity,
- $\ln(F)$ : log book debt, where book debt is short-term debt plus 50% of long-term debt,
- $1/\sigma_E$ : inverse equity volatility,
- $r_{i,t-1} - r_{m,t-1}$ : the firm's past-year excess stock return,
- $NI/TA$ : net income divided by total assets.

The regressions also include year and industry fixed effects, and standard errors are clustered by firm.

**Interpretation.** These controls absorb the usual structural and reduced-form determinants of default risk so that the CDS variables do not merely proxy for obvious balance-sheet weakness.

## 2.6 Instruments for CDS introduction

The two main instruments are:

- **Lender FX Hedging:** the average foreign-exchange hedging activity of the firm's recent lenders and bond underwriters, relative to assets.
- **Lender Tier 1 Capital:** the average Tier 1 capital ratio of the firm's recent lenders and underwriters.

Construction details:

- lenders come from DealScan;
- bond underwriters come from FISD;
- bank characteristics come from Federal Reserve call reports and Compustat Bank data;
- both instruments are averaged across banks that served the firm during the prior five years.

**Interpretation.** These instruments are meant to shift lenders' incentives to use CDS without directly changing the borrower's bankruptcy risk except through CDS activity.

## 2.7 Distress and creditor variables

For mechanism tests, the paper defines:

- **Financial distress:** a firm is distressed if its stock return lies in the bottom 5% of the market for two consecutive years; the firm is then classified as distressed in the third year. The paper also checks a 10% cutoff.
- **Number of Banks:** the number of unique bank lenders appearing in active syndicated-loan facilities over the prior five years.
- **$\Delta$ Number of Banks:** the change in lender count from one year before CDS inception to two years after.

**Interpretation.** These variables let the paper move from "CDS increase bankruptcy" to more specific channels: tougher creditors in distress and harder creditor coordination.

## 2.8 Descriptive facts that matter for interpretation

- The average monthly bankruptcy probability in the full sample is only 0.14%.
- The average monthly downgrade probability is 0.59%.
- Among the 901 CDS firms, 60 eventually file for bankruptcy, a rate of 6.7%.
- The median (mean) fraction of no-restructuring contracts among CDS firms is 0.61 (0.55).

**Interpretation.** Even small absolute marginal effects can be economically important because bankruptcy is a rare event.

### 3 Models and empirical specifications (all equations + interpretation)

#### 3.1 Economic mechanism and hypotheses

The paper starts from the idea that CDS alter the creditor-borrower relationship.

Potential channels pushing risk *up*:

- hedged creditors monitor less intensely;
- hedged creditors become tougher in debt renegotiation;
- distressed firms become harder to refinance or restructure;
- more lenders enter because CDS facilitate hedging, worsening coordination in workouts.

Potential channels pushing risk *down*:

- creditors can hedge, so they may lend more;
- better access to credit can improve financial flexibility;
- CDS prices may reveal information and strengthen discipline.

The four formal hypotheses are:

- **Hypothesis 1:** credit risk rises after CDS introduction.
- **Hypothesis 2:** conditional on financial distress, CDS firms are more likely to file for bankruptcy.
- **Hypothesis 3:** the rise in bankruptcy risk is larger when a greater share of contracts are no-restructuring CDS.
- **Hypothesis 4:** the number of lenders rises after CDS introduction, and bankruptcy risk rises with the number of lenders.

**Interpretation.** The paper is fundamentally a test of whether the empty-creditor mechanism shows up in firm-level default outcomes.

#### 3.2 Baseline discrete-time hazard/logit model

The core bankruptcy equation is a monthly discrete-time hazard model:

$$\Pr(Y_{it} = 1 \mid X_{i,t-1}) = \frac{1}{1 + \exp(-\alpha - \beta'X_{i,t-1})},$$

where:

- $Y_{it} = 1$  if firm  $i$  files for bankruptcy in month  $t$ ,
- $X_{i,t-1}$  contains the lagged firm controls plus the CDS variables,
- the key regressor is *CDS Active*.

The same logic is applied to rating downgrades.

**Interpretation.** The coefficient on *CDS Active* asks whether a firm’s downgrade or bankruptcy risk jumps after CDS begin trading, holding constant the usual determinants of credit quality.

### 3.3 Why *CDS Firm* and *CDS Active* both enter

The paper estimates specifications both with and without *CDS Firm*.

- **CDS Firm** captures time-invariant differences between firms that ever receive CDS and firms that never do.
- **CDS Active** captures the within-firm change after CDS inception.

**Interpretation.** This separation is important. It prevents the paper from attributing the fact that future CDS firms are different ex ante to the causal effect of CDS introduction itself.

### 3.4 Selection equation for CDS introduction

To model CDS introduction explicitly, the paper uses a probit selection equation:

$$CDSActive^* = \gamma_0 + \gamma_1 X + \gamma_2 Z + \nu,$$

with observed treatment

$$CDSActive = 1\{CDSActive^* > 0\}.$$

Here:

- $X$  includes firm size, leverage, ratings, profitability, tangibility, sales, working capital, retained earnings, cash, capital expenditures, debt structure, and lender controls;
- $Z$  contains the two lender-based instruments: lender FX hedging and lender Tier 1 capital.

**Interpretation.** This equation isolates which firms are likely to attract CDS coverage, and it creates the propensity scores used in the matching exercise.

### 3.5 Propensity-score matching

The paper matches each CDS firm to non-CDS firms with similar predicted probabilities of receiving CDS trading based on the selection model. It uses three variants:

- nearest one match,
- nearest one within a 1% propensity-score distance,
- nearest two matches.

The matched-sample bankruptcy regressions then re-estimate the effect of *CDS Active*.

**Interpretation.** Matching is treated as a robustness test: if the CDS effect survives in a sample of firms with similar observables, the raw result is less likely to be driven by obvious selection on size, ratings, or leverage.



### 3.6 Joint estimation with endogenous binary treatment

The paper then estimates a binary-outcome system with endogenous CDS introduction:

$$\begin{aligned}Bankruptcy^* &= \alpha + \beta CDSActive + \mu X + \varepsilon, \\Bankruptcy &= 1\{Bankruptcy^* > 0\}, \\CDSActive^* &= \gamma_0 + \gamma_1 X + \gamma_2 Z + \nu, \\CDSActive &= 1\{CDSActive^* > 0\}.\end{aligned}$$

The joint full-information maximum likelihood (FIML) estimator uses the correlation between  $\varepsilon$  and  $\nu$  to account for endogeneity.

**Interpretation.** If investors start CDS trading precisely when unobserved bankruptcy risk is rising, then  $\varepsilon$  and  $\nu$  are correlated. FIML tries to model that correlation directly rather than ignore it.

### 3.7 Mechanism regressions

After establishing the baseline CDS effect, the paper modifies the hazard/logit setup in four ways.

#### 1. Continuous CDS exposure.

- Replace or augment *CDS Active* with *CDS Notional Outstanding/Total Debt* and an instrumented version of it.

#### 2. Bankruptcy conditional on distress.

- Restrict the sample to firms already in severe financial distress and ask whether CDS predict bankruptcy within that distressed sample.

#### 3. Contract-type mechanism.

- Use *No Restructuring CDS* and instrumented NR exposure to test whether contracts that do not pay in restructuring are especially associated with bankruptcy.

#### 4. Creditor coordination.

- Regress the change in lender counts around CDS inception on *CDS Active*, and then add the number of banks or the change in number of banks into bankruptcy regressions.

**Interpretation.** These additional specifications are what turns the paper from a pure reduced-form study into a mechanism paper centered on empty creditors and coordination failure.

### 3.8 Standard errors and panel structure

- The main panel is monthly, 1997–2009.
- Regressions include year and industry fixed effects.
- Standard errors are clustered at the firm level because observations for the same firm are serially correlated over time.

**Interpretation.** The paper’s main identifying variation is within-firm timing relative to CDS inception, not cross-sectional differences alone.

## 4 Identification and instruments (what makes the estimates credible)

### 4.1 What identifies the baseline estimate

The baseline estimate is identified from the change in a firm's credit risk after CDS trading begins, controlling for:

- structural credit-risk covariates,
- year effects,
- industry effects,
- and, in some specifications, a level difference for ever-CDS firms through *CDS Firm*.

**Interpretation.** In the cleanest reading, the paper compares a firm's risk before and after CDS inception and asks whether that change is too large to be explained by ordinary default determinants.

### 4.2 Why endogeneity is the main concern

Two concerns are central:

- **Selection on observables:** larger, rated, better-known firms are more likely to be referenced in CDS markets.
- **Selection on unobservables / anticipation:** market participants may start CDS trading because they foresee future deterioration in credit quality.

**Interpretation.** The paper never claims the raw *CDS Active* coefficient is enough by itself. Most of the credibility comes from showing that the effect survives several different corrections for this problem.

### 4.3 Why the instruments are plausible

**Lender FX Hedging.** Banks and underwriters with heavier FX hedging activity are more active derivative users and are more likely to use CDS as well.

**Lender Tier 1 Capital.** Banks with lower capital ratios have a stronger incentive to hedge credit risk.

The authors argue that these lender characteristics should affect the firm's bankruptcy risk mainly through the likelihood that the firm's creditors use CDS.

**Interpretation.** The exclusion restriction is not perfect beyond question, but it is economically motivated: the instruments live on the lender side, whereas the outcome lives on the borrower side.

### 4.4 Evidence that the instruments matter

The first-stage CDS selection results show:

- Lender FX Hedging enters positively and strongly.
- Lender Tier 1 Capital enters negatively and strongly.
- Instrument statistics are economically and statistically strong.
- The Sargan overidentification test fails to reject exogeneity of the instruments.

**Interpretation.** The instruments do shift CDS trading in the expected direction, which is necessary before the second-stage interpretation is taken seriously.

## 4.5 Why the matching and FIML results matter

Propensity-score matching answers:

- Does the CDS effect survive if we compare treated firms to observationally similar untreated firms?

FIML answers:

- Does the CDS effect survive if we jointly model bankruptcy and CDS introduction as correlated binary outcomes?

**Interpretation.** These are complementary approaches. Matching reduces dependence on functional form but only addresses observables; FIML is stronger econometrically but relies on correct parametric specification.

## 4.6 Alternative explanations the paper investigates

The paper explicitly considers and partly rules out the following alternatives:

- CDS just proxy for general firm deterioration that already existed before the contracts were introduced.
- The results come only from crisis years or a narrow part of the sample.
- The coefficients are driven only by a binary regime shift rather than actual CDS exposure.
- The result is unrelated to creditor toughness and instead only reflects leverage changes.

**Interpretation.** The paper does not deny that leverage may also change after CDS introduction; instead, it argues that even after taking that possibility seriously, the evidence still points to a distinct empty-creditor and coordination channel.

## 4.7 What the paper can and cannot distinguish

What the paper does well:

- It shows that CDS inception predicts worse ex post credit outcomes.
- It shows that this is strongest in precisely the settings highlighted by the empty-creditor theory.

What it cannot pin down perfectly:

- the exact net welfare effect of CDS markets,
- the exact decomposition between tougher renegotiation, reduced monitoring, rollover risk, and leverage,
- and the identity or net CDS position of the specific creditors who become “empty.”

**Interpretation.** The paper is strongest on the existence of a real CDS effect on credit risk and on the qualitative consistency with empty-creditor theory; it is weaker on exact welfare accounting and micro-level creditor positions.

## 4.8 Relation to the literature

The paper sits at the intersection of:

- the corporate default and bankruptcy literature,
- the law-and-finance literature on creditor rights,
- and the emerging literature on how derivatives change real incentives rather than just prices.

It complements:

- Ashcraft and Santos (2009) on the cost of debt,
- Saretto and Tookes (2013) on leverage and maturity,
- and Bolton and Oehmke (2011) on empty-creditor theory.

## 5 Tables and figures

### 5.1 Table 1 and Figure 1: sample structure and rating deterioration

**Read:**

- Table 1 shows that the paper starts from 901 CDS firms and 1,628 bankruptcy filings over 1997–2009.
- CDS introductions are concentrated in 2000–2003.
- Among CDS firms, 60 eventually file for bankruptcy.
- Figure 1 shows a visible shift in ratings after CDS inception: the mass moves away from A and AA and toward junk-rated and unrated categories.

**Economic message.** Before any regression is run, the basic descriptive evidence already points in the same direction as the paper’s hypothesis: credit quality looks worse after CDS begin trading.

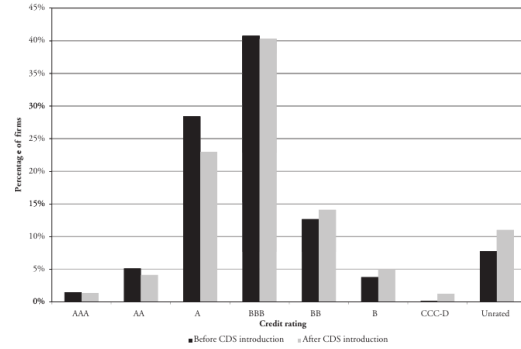
**Table 1**  
Credit default swaps trading and bankruptcies by year

| (1)               | (2)              | (3)               | (4)                | (5)                   | (6)                   | (7)                       |
|-------------------|------------------|-------------------|--------------------|-----------------------|-----------------------|---------------------------|
| Year              | Total # of firms | # of bankruptcies | # of new CDS firms | # of active CDS firms | # of CDS bankruptcies | # of non-CDS bankruptcies |
| 1997*             | 9366             | 50                | 22                 | 22                    | 0                     | 50                        |
| 1998              | 9546             | 92                | 58                 | 72                    | 0                     | 92                        |
| 1999              | 9545             | 118               | 55                 | 106                   | 0                     | 118                       |
| 2000              | 9163             | 158               | 102                | 196                   | 1                     | 157                       |
| 2001              | 8601             | 257               | 172                | 334                   | 8                     | 249                       |
| 2002              | 8190             | 225               | 221                | 547                   | 12                    | 213                       |
| 2003              | 7876             | 156               | 93                 | 582                   | 5                     | 151                       |
| 2004              | 7560             | 86                | 58                 | 593                   | 0                     | 86                        |
| 2005              | 7318             | 76                | 73                 | 629                   | 5                     | 71                        |
| 2006              | 6993             | 49                | 28                 | 533                   | 2                     | 47                        |
| 2007              | 6651             | 61                | 9                  | 418                   | 1                     | 60                        |
| 2008              | 6223             | 121               | 9                  | 375                   | 4                     | 117                       |
| 2009 <sup>†</sup> | 5686             | 179               | 1                  | 234                   | 22                    | 157                       |
| Total             |                  | 1628              | 901                |                       | 60                    | 1568                      |
| Final sample      |                  | 57.7%             | 61.6%              |                       | 70.0%                 | 57.3%                     |
| Matched sample    |                  | 3.1%              | 56.7%              |                       | 56.7%                 | 1.0%                      |
| Matched final     |                  | 2.8%              | 53.3%              |                       | 51.7%                 | 0.9%                      |

This table reports the distribution of firms, including those with credit default swaps (CDS) traded, and bankruptcy events, by year, in our sample, between 1997 and 2009. The sample of all firms is drawn from Compustat, and includes all companies in the database from 1997–2009. The CDS data are taken from CreditTrade and the GFI Group. There are 901 firms in our sample that have CDS traded at some point during the sample period of June 1997 to April 2009. The bankruptcy data are obtained from New Generation Research's "Public and Major Company Database," the UCLA-LePucki Bankruptcy Research Database (BRD), the Altman-NYU Salomon Center Bankruptcy List, the Fixed Income Securities Database (FISD), and Moody's Annual Reports on Bankruptcy and Recovery. The combined database includes all public companies that filed for bankruptcy during the period; it also includes selected private firms that are deemed significant. The first column in the table is the year. The second column in the table shows the total number of U.S. companies included in the Compustat database. The third column shows the number of bankruptcies in the year. The fourth column reports the number of firms for which CDS trading was initiated during the year in question. The fifth column presents firms with active CDS trading during each year. The last two columns report the number of CDS firms that filed for bankruptcy and the number of non-CDS firms that filed for bankruptcy, respectively. The last three rows report the number of bankruptcies (or firms) in the final sample, the propensity score-matched sample, and the propensity score-matched final sample, as percentages of the raw sample. (\* from June 1997; † until April 2009)

of firms with *Active CDS* trading firms across the years, respectively. More CDS contracts were introduced in the 2000–2003 period than in any earlier or later periods. Among the 901 distinct CDS trading firms, 60 (6.7%) subsequently

(a) Table 1

**Figure 1**  
Rating distribution around the introduction of credit default swaps

This figure plots the credit rating distributions for firms with credit default swaps (CDS), before the inception of CDS trading and two years after the inception of CDS trading. The credit ratings are taken from S&P Credit Ratings. The CDS data come from CreditTrade and the GFI Group. There are 901 firms in our sample that have CDS traded at some point during the sample period of June 1997 to April 2009.

using the Kolmogorov-Smirnov test.<sup>8</sup> We use the one-sided test for stochastic dominance. The  $D$ -statistic of the one-sided Kolmogorov-Smirnov test (the maximum difference between cumulative distributions) is significant at the 1% level, indicating that the credit rating distribution shifts to the right (lower rating quality) after CDS trading commences. In other words, the ratings before CDS trading is introduced stochastically dominate the ratings afterward.

(b) Figure 1

**Figure 1:** Sample structure and the rating distribution around CDS inception.

## 5.2 Table 2: baseline downgrade and bankruptcy effects

Read:

- In the downgrade regressions, *CDS Active* is positive and highly significant.
- In the bankruptcy regression without *CDS Firm*, the coefficient on *CDS Active* is 2.373 with an odds ratio of 10.73 and a marginal effect of 0.33 percentage points.
- Since the sample bankruptcy probability is only 0.14%, this absolute effect is economically very large.
- With *CDS Firm* included, *CDS Active* remains positive and significant, though smaller, which suggests the result is not just coming from cross-sectional differences between future CDS and non-CDS firms.

**Economic message.** Table 2 is the paper's core reduced-form result: a firm's credit risk rises after CDS trading begins.

**Table 2**  
Impact of credit default swaps trading on credit quality

|                                                   | Probability of downgrade |                      | Probability of bankruptcy |                      |
|---------------------------------------------------|--------------------------|----------------------|---------------------------|----------------------|
|                                                   | (1)                      | (2)                  | (3)                       | (4)                  |
| $\ln(E)$                                          | -0.735***<br>(0.022)     | -0.736***<br>(0.022) | -0.713***<br>(0.026)      | -0.710***<br>(0.026) |
| $\ln(F)$                                          | 0.507***<br>(0.024)      | 0.503***<br>(0.024)  | 0.711***<br>(0.026)       | 0.713***<br>(0.026)  |
| $1/\sigma_E$                                      | -0.062*<br>(0.035)       | -0.017<br>(0.032)    | -1.626***<br>(0.251)      | -1.675***<br>(0.251) |
| $r_{it-1} - r_{mt-1}$                             | -0.281***<br>(0.061)     | -0.252***<br>(0.059) | -1.320***<br>(0.192)      | -1.331***<br>(0.192) |
| $NI/TA$                                           | -0.003<br>(0.008)        | -0.000<br>(0.008)    | -0.038***<br>(0.009)      | -0.038***<br>(0.009) |
| <i>CDS Firm</i>                                   | 0.755***<br>(0.088)      |                      | -2.009***<br>(0.720)      |                      |
| <i>CDS Active</i>                                 | 0.691***<br>(0.096)      | 1.371***<br>(0.067)  | 2.373***<br>(0.736)       | 0.400**<br>(0.179)   |
| <i>CDS Active Odds Ratio</i>                      | 1.925                    | 3.939                | 10.730                    | 1.492                |
| <i>CDS Active Marginal Effect</i>                 | 0.39%                    | 0.78%                | 0.33%                     | 0.06%                |
| Time fixed effects                                | yes                      | yes                  | yes                       | yes                  |
| Industry fixed effects                            | yes                      | yes                  | yes                       | yes                  |
| Clustered standard error                          | yes                      | yes                  | yes                       | yes                  |
| Pseudo $R^2$                                      | 15.08%                   | 14.75%               | 24.18%                    | 24.06%               |
| N                                                 | 658,966                  | 658,966              | 658,966                   | 658,966              |
| % of CDS observations                             | 10.83%                   | 10.83%               | 10.83%                    | 10.83%               |
| % of CDS downgrades (bankruptcies)                | 34.74%                   | 34.74%               | 4.47%                     | 4.47%                |
| Sample probability of<br>a downgrade (bankruptcy) | 0.59%                    | 0.59%                | 0.14%                     | 0.14%                |

This table presents the estimates of the probabilities of credit downgrades and bankruptcy, using a logistic model in a sample including firms with credit default swaps (CDS) and all non-CDS firms.  $\ln(E)$  is the logarithm of the firm's market value of equity.  $\ln(F)$  is the logarithm of the book value of the firm's debt, where book debt is the sum of short-term debt and 50% of long-term debt.  $1/\sigma_E$  is the inverse of the firm's annualized equity volatility.  $r_{it-1} - r_{mt-1}$  is the firm's excess return over the past year, and  $NI/TA$  is the firm's ratio of net income to total assets. To estimate the impact of CDS trading on the probability of a credit downgrade or bankruptcy, we include CDS variables in the model specification. *CDS Firm* equals one if the firm is in the CDS sample and zero otherwise. *CDS Active* is a dummy variable that equals one after the inception of CDS trading and zero before CDS trading. The coefficient of interest is that of *CDS Active*, which captures the impact of CDS trading on the probability of a credit downgrade or bankruptcy after the inception of CDS trading. The sample period is 1997–2009, based on monthly observations. (\*\*\*) significant at the 1% level, \*\* significant at the 5% level, and \* significant at the 10% level. The numbers in parentheses are standard errors.)

**Figure 2:** Table 2 from the paper: baseline effect of CDS trading on downgrades and bankruptcy.

### 5.3 Table 3: what predicts CDS trading?

**Read:**

- Larger firms are much more likely to have CDS traded on them.
- CDS are more likely on highly levered but relatively high-quality, rated firms.
- Profitability, tangibility, and working capital enter positively.
- Lender FX Hedging enters positively and Lender Tier 1 Capital enters negatively, both strongly and with pseudo- $R^2$  near 39%.

**Economic message.** CDS introduction is clearly selective, which is exactly why the later matching and FIML corrections are essential.

**Table 3**  
Probability of credit default swaps trading

|                                  | Probability of CDS trading |                           |                           |
|----------------------------------|----------------------------|---------------------------|---------------------------|
|                                  | CDS prediction<br>model 1  | CDS prediction<br>model 2 | CDS prediction<br>model 3 |
| <i>Ln(Assets)</i>                | 0.790***<br>(0.006)        | 0.804***<br>(0.006)       | 0.797***<br>(0.006)       |
| <i>Leverage</i>                  | 0.429***<br>(0.025)        | 0.440***<br>(0.025)       | 0.431***<br>(0.026)       |
| <i>ROA</i>                       | -0.001<br>(0.001)          | -0.001<br>(0.001)         | -0.001<br>(0.001)         |
| $r_{it-1} - r_{mt-1}$            | -0.104***<br>(0.011)       | -0.104***<br>(0.011)      | -0.104***<br>(0.011)      |
| <i>Equity Volatility</i>         | 0.063***<br>(0.017)        | 0.069***<br>(0.017)       | 0.067***<br>(0.017)       |
| <i>PPENT/Total Asset</i>         | 0.306***<br>(0.031)        | 0.321***<br>(0.031)       | 0.307***<br>(0.031)       |
| <i>Sales/Total Asset</i>         | -0.026***<br>(0.009)       | -0.027***<br>(0.003)      | -0.026***<br>(0.003)      |
| <i>EBIT/Total Asset</i>          | 0.315***<br>(0.064)        | 0.375***<br>(0.064)       | 0.338***<br>(0.064)       |
| <i>WCAP/Total Asset</i>          | 0.142***<br>(0.024)        | 0.145***<br>(0.024)       | 0.143***<br>(0.024)       |
| <i>RE/Total Asset</i>            | 0.022***<br>(0.005)        | 0.023***<br>(0.005)       | 0.024***<br>(0.005)       |
| <i>Cash/Total Asset</i>          | 0.290***<br>(0.023)        | 0.302***<br>(0.023)       | 0.294***<br>(0.023)       |
| <i>CAPX/Total Asset</i>          | -1.611***<br>(0.122)       | -1.677***<br>(0.122)      | -1.604***<br>(0.122)      |
| <i>Rated</i>                     | 0.667***<br>(0.203)        | 0.645***<br>(0.205)       | 0.638***<br>(0.205)       |
| <i>Senior Unsecured Debt</i>     | 0.375***<br>(0.014)        | 0.377***<br>(0.014)       | 0.375***<br>(0.014)       |
| <i>Lender Size</i>               | 0.369***<br>(0.011)        | 0.378***<br>(0.011)       | 0.385***<br>(0.011)       |
| <i>Lender Credit Derivatives</i> | 1.006***<br>(0.024)        | 1.013***<br>(0.024)       | 1.019***<br>(0.025)       |
| <i>Lender FX Hedging</i>         | 8.979***<br>(0.788)        |                           | 9.104***<br>(0.789)       |
| <i>Lender Tier 1 Capital</i>     |                            | -3.865***<br>(0.756)      | -4.000***<br>(0.757)      |
| F-statistic (instruments)        | 129.89                     | 26.13                     | 159.74                    |
| p-value (F-statistic)            | 0.000                      | 0.000                     | 0.000                     |
| Credit rating controls           | yes                        | yes                       | yes                       |
| Time fixed effects               | yes                        | yes                       | yes                       |
| Industry fixed effects           | yes                        | yes                       | yes                       |
| Clustered standard error         | yes                        | yes                       | yes                       |
| Pseudo $R^2$                     | 38.96%                     | 38.79%                    | 38.99%                    |
| N                                | 690,111                    | 690,111                   | 690,111                   |
| % of CDS observations            | 6.15%                      | 6.15%                     | 6.15%                     |

This table presents the estimates of the probability of credit default swaps (CDS) trading, obtained using a probit model. Propensity scores are estimated based on the model parameters. *Ln(Assets)* is the logarithm of the firm's total asset value. *Leverage* is defined as the ratio of book debt to the sum of book debt and market equity, where

**Figure 3:** Table 3 from the paper: the first-stage / propensity model for CDS trading.

## 5.4 Tables 4 and 5: endogeneity corrections

**Read:**

- In Table 4, *CDS Active* remains positive and significant in all three propensity-score-matched samples; the odds ratios range from roughly 5.9 to 9.6.
- In Table 5, the joint FIML estimates also keep *CDS Active* positive and significant.
- The endogeneity test in Table 5 rejects exogeneity of CDS trading, so modeling selection is not cosmetic; it matters.

**Economic message.** The CDS effect survives both observable-selection correction and joint treatment-outcome estimation. That is the paper's main credibility step.

**Table 4**  
Credit default swaps trading and probability of bankruptcy: Propensity score matching

|                                    | Probability of bankruptcy |                          |                      |
|------------------------------------|---------------------------|--------------------------|----------------------|
|                                    | Nearest one matching      | Nearest one PS diff. <1% | Nearest two matching |
| $\ln(E)$                           | -1.206***<br>(0.246)      | -1.350***<br>(0.309)     | -0.993***<br>(0.195) |
| $\ln(F)$                           | 1.274***<br>(0.211)       | 1.319***<br>(0.248)      | 1.008***<br>(0.203)  |
| $1/\sigma_E$                       | -0.078<br>(1.043)         | 0.048<br>(0.904)         | -0.612<br>(1.208)    |
| $r_{it-1} - r_{mt-1}$              | -2.054*<br>(1.190)        | -1.655<br>(1.046)        | -1.59<br>(1.290)     |
| $N/TA$                             | -0.026<br>(0.106)         | -2.919***<br>(0.753)     | 0.034<br>(0.095)     |
| $CDS\ Firm$                        | -0.262<br>(1.050)         | -0.279<br>(1.261)        | -0.555<br>(0.879)    |
| $CDS\ Active$                      | 1.858**<br>(0.910)        | 2.261**<br>(1.082)       | 1.771**<br>(0.802)   |
| $CDS\ Active\ Odds\ Ratio$         | 6.411                     | 9.593                    | 5.877                |
| $CDS\ Active\ Marginal\ Effect$    | 0.07%                     | 0.07%                    | 0.07%                |
| Time fixed effects                 | yes                       | yes                      | yes                  |
| Industry fixed effects             | yes                       | yes                      | yes                  |
| Clustered standard error           | yes                       | yes                      | yes                  |
| Pseudo $R^2$                       | 37.29%                    | 38.53%                   | 36.91%               |
| N                                  | 119,710                   | 108,401                  | 168,600              |
| % of CDS observations              | 57.49%                    | 56.58%                   | 40.82%               |
| % of CDS bankruptcies              | 68.89%                    | 71.43%                   | 49.21%               |
| Sample probability of a bankruptcy | 0.03%                     | 0.03%                    | 0.03%                |

This table presents the estimates of the probability of bankruptcy, using a logistic model in a sample including firms with credit default swaps (CDS) and non-CDS propensity score-matched firms. Propensity score-matched firms are selected based on propensity scores estimated from the model of the probability of CDS trading presented in Table 3.  $\ln(E)$  is the logarithm of the firm's market value of equity.  $\ln(F)$  is the logarithm of the book value of the firm's debt, where book debt is the sum of short-term debt and 50% of long-term debt.  $1/\sigma_E$  is the inverse of the firm's annualized equity volatility.  $r_{it-1} - r_{mt-1}$  is the firm's excess return over the past year, and  $N/TA$  is the firm's ratio of net income to total assets. To estimate the impact of CDS trading on the probability of bankruptcy, we include CDS variables in the model specification.  $CDS\ Firm$  equals one if the firm is in the CDS sample and zero otherwise.  $CDS\ Active$  is a dummy variable that equals one after the inception of CDS trading and zero before CDS trading begins. The coefficient of interest is  $CDS\ Active$ , which captures the impact of the inception of CDS trading on the probability of bankruptcy. The second column presents the analysis in the baseline-matched sample, that is the "nearest one" propensity score-matched firms, selected based on the CDS prediction model 3 in Table 3. The third column presents the same analysis, but for the "nearest one" with a propensity score difference within 1%. The fourth column uses the two matching firms with the nearest propensity scores. The sample period is 1997–2009, based on monthly observations. (\*\* significant at the 1% level, \* significant at the 10% level, and \*\*\* significant at the 5% level.)

(a) Table 4

**Table 5**  
Credit default swaps trading and probability of bankruptcy: Full-information maximum likelihood

|                          | Probability of bankruptcy |                      |
|--------------------------|---------------------------|----------------------|
|                          | (1)                       | (2)                  |
| $\ln(E)$                 | -0.581***<br>(0.022)      | -0.580***<br>(0.022) |
| $\ln(F)$                 | 0.628***<br>(0.024)       | 0.630***<br>(0.024)  |
| $1/\sigma_E$             | -0.284***<br>(0.044)      | -0.298***<br>(0.044) |
| $r_{it-1} - r_{mt-1}$    | -0.072**<br>(0.028)       | -0.074***<br>(0.028) |
| $N/TA$                   | -0.288***<br>(0.031)      | -0.288***<br>(0.031) |
| $CDS\ Firm$              | -0.340**<br>(0.147)       |                      |
| $CDS\ Active$            | 0.864***<br>(0.175)       | 0.552***<br>(0.111)  |
| Time fixed effects       | yes                       | yes                  |
| Industry fixed effects   | yes                       | yes                  |
| Clustered standard error | yes                       | yes                  |
| N                        | 558,376                   | 558,376              |
| % of CDS observations    | 11.98%                    | 11.98%               |
| % of CDS bankruptcies    | 4.47%                     | 4.47%                |
| Endogeneity test         |                           |                      |
| Rho                      | 0.038***                  | 0.035***             |

This table presents the full-information maximum likelihood (FIML) results of the impact of credit default swaps (CDS) on a firm's probability of bankruptcy.  $\ln(E)$  is the logarithm of the firm's market value of equity.  $\ln(F)$  is the logarithm of the book value of the firm's debt, where book debt is the sum of short-term debt and 50% of long-term debt.  $1/\sigma_E$  is the inverse of the firm's annualized equity volatility.  $r_{it-1} - r_{mt-1}$  is the firm's excess return over the past year, and  $N/TA$  is the firm's ratio of net income to total assets. To estimate the impact of CDS trading on the probability of bankruptcy, we include CDS variables in the model specification.  $CDS\ Firm$  equals one if the firm is in the CDS sample and zero otherwise.  $CDS\ Active$  is a dummy variable that equals one after the inception of CDS trading and zero before CDS trading begins. The coefficient of interest is that of  $CDS\ Active$ , which captures the impact of the inception of CDS trading on the probability of bankruptcy. The sample period is 1997–2009, based on monthly observations. (\*\* significant at the 1% level, \* significant at the 5% level, and \*\*\* significant at the 10% level. The numbers in parentheses are standard errors.)

A17 of the Internet Appendix The coefficient estimates for  $CDS\ Active$  are

(b) Table 5

**Figure 4:** Endogeneity corrections via propensity-score matching and FIML.

## 5.5 Tables 6 and 7: CDS intensity and bankruptcy in distress

Read:

- Table 6 shows that bankruptcy risk rises with *CDS Notional Outstanding/Total Debt*; the result survives instrumentation of the notional measure.
- Table 7 shows that within financially distressed firms, *CDS Active* raises bankruptcy risk sharply. In the main distressed sample, the marginal effect is 19.84% against a base default rate of 17.71%.

**Economic message.** These two tables move beyond the headline “CDS matters” result. More CDS exposure matters more, and it matters especially when renegotiation is most likely to become pivotal: when the borrower is already in distress.



**Table 6**  
Credit default swaps outstanding and probability of bankruptcy

|                                                        | Probability of bankruptcy |                      |
|--------------------------------------------------------|---------------------------|----------------------|
|                                                        | (1)                       | (2)                  |
| $\ln(E)$                                               | -0.689***<br>(0.028)      | -0.674***<br>(0.083) |
| $\ln(F)$                                               | 0.652***<br>(0.028)       | 0.631***<br>(0.085)  |
| $1/\sigma_E$                                           | -1.533***<br>(0.121)      | -2.027***<br>(0.371) |
| $r_{it-1} - r_{mt-1}$                                  | -0.620***<br>(0.102)      | -0.830***<br>(0.237) |
| $NI/TA$                                                | -0.076***<br>(0.025)      | -0.111***<br>(0.023) |
| $CDS_{Firm}$                                           | -0.582***<br>(0.206)      | -0.487<br>(0.477)    |
| $CDS_{Notional\ Outstanding/Total\ Debt}$              | 0.071**<br>(0.030)        |                      |
| $Instrumented\ CDS\ Notional\ Outstanding/Total\ Debt$ |                           | 0.946***<br>(0.213)  |
| Time fixed effects                                     | yes                       | yes                  |
| Industry fixed effects                                 | yes                       | yes                  |
| Clustered standard error                               | yes                       | yes                  |
| Pseudo $R^2$                                           | 15.82%                    | 15.25%               |
| N                                                      | 658,966                   | 658,966              |
| % of CDS observations                                  | 10.83%                    | 10.83%               |
| % of CDS bankruptcies                                  | 4.47%                     | 4.47%                |
| Sample probability of a bankruptcy                     | 0.14%                     | 0.14%                |

This table shows the impact of credit default swaps (CDS) on a firm's probability of bankruptcy.  $\ln(E)$  is the logarithm of the firm's market value of equity.  $\ln(F)$  is the logarithm of the book value of the firm's debt, where book debt is the sum of short-term debt and 50% of long-term debt.  $1/\sigma_E$  is the inverse of the firm's annualized equity volatility.  $r_{it-1} - r_{mt-1}$  is the firm's excess return over the past year, and  $NI/TA$  is the firm's ratio of net income to total assets.  $CDS_{Firm}$  equals one if the firm has CDS trading at any point in time and zero otherwise. The CDS impact is measured as the total notional CDS outstanding, scaled by the book value of the total debt ( $CDS_{Notional\ Outstanding/Total\ Debt}$ ) or the *Instrumented CDS Notional Outstanding/Total Debt*. The sample period is 1997–2009, based on monthly observations. (\*\*\*) significant at the 1% level, \*\* significant at the 5% level, and \* significant at the 10% level. The numbers in parentheses are standard errors.)

essentially for hedging purposes. We then use the *Instrumented CDS Marginal Effect*

(a) Table 6

**Table 7**  
Bankruptcy prediction for distressed firms

|                                    | Probability of bankruptcy |                      |                      |                      |
|------------------------------------|---------------------------|----------------------|----------------------|----------------------|
|                                    | (1)                       | (2)                  | (3)                  | (4)                  |
| $\ln(E)$                           | -0.492***<br>(0.089)      | -0.398***<br>(0.077) | -0.495***<br>(0.103) | -0.430***<br>(0.093) |
| $\ln(F)$                           | 0.504***<br>(0.089)       | 0.461***<br>(0.071)  | 0.506***<br>(0.103)  | 0.457***<br>(0.092)  |
| $1/\sigma_E$                       | -0.353<br>(0.532)         | -1.072***<br>(0.376) | 0.071<br>(0.654)     | -0.462<br>(0.542)    |
| $r_{it-1} - r_{mt-1}$              | -0.074<br>(0.224)         | -0.085<br>(0.088)    | -0.076<br>(0.271)    | -0.100<br>(0.223)    |
| $NI/TA$                            | -0.046<br>(0.033)         | 0.002<br>(0.026)     | -0.049<br>(0.033)    | -0.043<br>(0.033)    |
| <i>Senior Secured Debt</i>         |                           |                      |                      | 0.273<br>(3.648)     |
| <i>Both Bond and Loan</i>          |                           |                      |                      | 0.696**<br>(0.323)   |
| <i>PPENT/Total Asset</i>           |                           |                      |                      | 0.422<br>(0.643)     |
| $CDS_{Active}$                     | 1.626**<br>(0.663)        | 1.064**<br>(0.484)   | 1.655**<br>(0.821)   | 1.368**<br>(0.671)   |
| $CDS_{Active\ Odds\ Ratio}$        | 5.083                     | 2.898                | 5.233                | 3.927                |
| $CDS_{Active\ Marginal\ Effect}$   | 19.84%                    | 9.99%                | 15.66%               | 16.51%               |
| Time fixed effects                 | yes                       | yes                  | yes                  | yes                  |
| Industry fixed effects             | yes                       | yes                  | yes                  | yes                  |
| Pseudo $R^2$                       | 23.24%                    | 19.91%               | 31.31%               | 24.33%               |
| N                                  | 655                       | 1,229                | 625                  | 655                  |
| % of CDS observations              | 2.29%                     | 3.25%                | 1.92%                | 2.29%                |
| % of CDS bankruptcies              | 6.11%                     | 5.96%                | 5.81%                | 6.11%                |
| Sample probability of a bankruptcy | 17.71%                    | 12.29%               | 13.76%               | 17.71%               |

This table shows the results of a bankruptcy analysis conducted in a sample of distressed firms with and without credit default swaps (CDS) traded. The information on the firms at the time of distress is used to predict the probability of bankruptcy. Distressed firms are defined as firms in the bottom 5% (models 1, 3, 4) or 10% (model 2) of stock returns in two consecutive years, following [Gibson, John, and Lamm \(1994\)](#) and [Demireu and James \(2013\)](#). Model 3 excludes crisis periods from the prediction. Model 4 includes additional controls.  $\ln(E)$  is the logarithm of the firm's market value of equity.  $\ln(F)$  is the logarithm of the book value of the firm's debt, where book debt is the sum of short-term debt and 50% of long-term debt.  $1/\sigma_E$  is the inverse of the firm's annualized equity volatility.  $r_{it-1} - r_{mt-1}$  is the firm's excess return over the past year, and  $NI/TA$  is the firm's ratio of net income to total assets. *Senior Secured Debt* is the ratio of senior secured debt to total debt. *Both Bond and Loan* is a dummy variable that equals one if the firm has both bonds and loans. *PPENT/Total Asset* is the ratio of

(b) Table 7

**Figure 5:** CDS exposure intensity and bankruptcy conditional on distress.

## 5.6 Figure 2 and Table 8: the no-restructuring channel

Read:

- Figure 2 shows that most contracts are modified restructuring (MR), but no-restructuring (NR) contracts become much more important later in the sample, especially after 2003.
- Table 8 shows that the share of NR contracts is positively and significantly related to bankruptcy risk.
- The result also survives an instrument based on the post-January-2003 shift toward NR contracts and remains significant inside the CDS sample itself.

**Economic message.** This is the sharpest mechanism test in the paper. If bankruptcy is especially more likely when CDS do *not* pay upon restructuring, that is exactly what the empty-creditor story predicts.

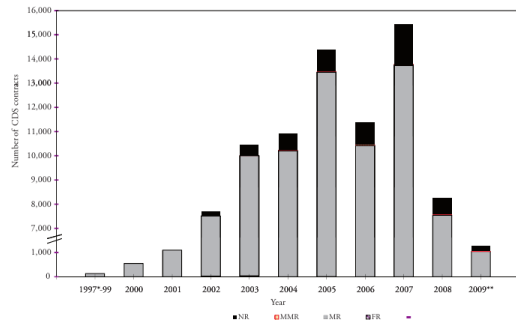


Figure 2

**Credit default swaps' restructuring clauses by year**

This figure plots the distribution of credit default swaps' (CDS') restructuring clauses, by year, in our sample, between 1997 and 2009. The CDS data are taken from CreditTrade and the GFI Group. There are four types of contract terms related to restructuring: full restructuring (FR), modified restructuring (MR), modified-modified restructuring (MMR), and no restructuring (NR). For firms with NR in the restructuring clause, the credit events do not include restructuring, whereas for the other types, they do. MR and MMR contracts impose restrictions on the type of bond that can be delivered in the event of default.

as a trigger event, and any debt obligation with a maturity of up to 30 years can be delivered in that event. Any restructuring is considered a credit event under MR as well; however, in this type of contract, deliverable obligations are limited to those with maturities within 30 months of the CDS contract's maturity. For

(a) Figure 2

Table 8

**Restructuring clauses of credit default swaps and probability of bankruptcy**

|                                    | Probability of bankruptcy |                      |                      |                      |                      |
|------------------------------------|---------------------------|----------------------|----------------------|----------------------|----------------------|
|                                    | (1)                       | (2)                  | (3)                  | (4)                  | (5)                  |
| $\ln(E)$                           | -0.716***<br>(0.026)      | -0.713***<br>(0.026) | -0.714***<br>(0.026) | -0.710***<br>(0.026) | -1.379***<br>(0.160) |
| $\ln(F)$                           | 0.715***<br>(0.026)       | 0.715***<br>(0.026)  | 0.713***<br>(0.026)  | 0.713***<br>(0.026)  | 1.381***<br>(0.157)  |
| $1/\alpha_E$                       | -1.636***<br>(0.254)      | -1.656***<br>(0.250) | -1.642***<br>(0.253) | -1.678***<br>(0.250) | 0.200<br>(0.330)     |
| $r_{it-1} - r_{mt-1}$              | -1.327***<br>(0.192)      | -1.330***<br>(0.193) | -1.330***<br>(0.193) | -1.335***<br>(0.193) | -2.126***<br>(0.708) |
| $NI/TA$                            | -0.037***<br>(0.009)      | -0.037***<br>(0.009) | -0.037***<br>(0.009) | -0.037***<br>(0.009) | -2.554***<br>(0.658) |
| <i>CDS Firm</i>                    | -0.206<br>(0.187)         | -0.609**<br>(0.277)  |                      |                      |                      |
| <i>No Restructuring CDS</i>        | 1.315**<br>(0.517)        | 1.023**<br>(0.438)   |                      |                      |                      |
| <i>Instrumented NR CDS</i>         |                           |                      | 0.658***<br>(0.192)  | 0.324***<br>(0.110)  | 0.776**<br>(0.363)   |
| Time fixed effects                 | yes                       | yes                  | yes                  | yes                  | yes                  |
| Industry fixed effects             | yes                       | yes                  | yes                  | yes                  | yes                  |
| Clustered standard error           | yes                       | yes                  | yes                  | yes                  | yes                  |
| Pseudo $R^2$                       | 24.06%                    | 24.05%               | 24.11%               | 24.07%               | 39.67%               |
| N                                  | 658,966                   | 658,966              | 658,966              | 658,966              | 71,366               |
| % of CDS observations              | 10.83%                    | 10.83%               | 10.83%               | 10.83%               | 100%                 |
| % of CDS bankruptcies              | 4.47%                     | 4.47%                | 4.47%                | 4.47%                | 100%                 |
| Sample probability of a bankruptcy | 0.14%                     | 0.14%                | 0.14%                | 0.14%                | 0.06%                |

This table shows the results from our investigation of the impact of the restructuring clauses of credit default swaps (CDS) on the probability of bankruptcy of the firms. The CDS impact is expected to be more significant for firms with more contracts in which "no restructuring" is specified as the restructuring clause. In models 1 and 2, for each CDS firm, we include a variable *No Restructuring CDS*, which is the total amount of active CDS contracts with "no restructuring" as the restructuring clause, scaled by the total number of CDS contracts trading on the firm. In models 3-5, for each CDS firm, we include *Instrumented NR CDS*, which is the instrumented *No Restructuring CDS* from the model predicting the amount of "no restructuring" CDS contracts. Models 1-4 investigate the impact of *No Restructuring CDS* in a sample including firms with and without CDS traded. Model 5 conducts the analysis in the CDS sample. *CDS Firm* equals one if the firm has CDS trading on it at any point in time and zero otherwise. The coefficients of interest are those of *No Restructuring CDS* and *Instrumented NR CDS*, which capture the impact of CDS.  $\ln(E)$  is the logarithm of the firm's market value of equity.  $\ln(F)$  is the logarithm of the book value of the firm's debt where book debt is the sum of short-term debt and 50% of long-

(b) Table 8

**Figure 6:** Evolution of restructuring clauses and their link to bankruptcy risk.

## 5.7 Table 9: creditor coordination and the lender base

**Read:**

- Panel A shows that CDS introduction increases the number of lenders. The event-style discussion in the text also reports about 1.4 extra lenders after one year and about 3 after two years relative to matched controls.
- In the cross-sectional specification, CDS firms have about 2.4 more lenders than matched non-CDS firms two years after CDS introduction.
- Panel B shows that bankruptcy risk rises with the number of banks and with the increase in the number of banks around CDS inception.

**Economic message.** CDS do not only change incentives of a given creditor; they also appear to change the creditor *structure* of the firm, which can make workouts harder and bankruptcy more likely.

**Table 9**  
**Credit default swaps trading, bank relationships, and probability of bankruptcy**

| Panel A: CDS and bank relationships |                     | Panel B: Bank relationships and bankruptcy risk |                      |                      |
|-------------------------------------|---------------------|-------------------------------------------------|----------------------|----------------------|
| $\Delta$ Number of banks            |                     | Probability of bankruptcy                       |                      |                      |
|                                     |                     | (1)                                             | (2)                  |                      |
| $\Delta \ln(\text{Asset})$          | 6.291***<br>(1.849) | $\ln(E)$                                        | -0.669***<br>(0.029) | -1.299***<br>(0.324) |
| $\Delta \text{ROA}$                 | -0.396<br>(2.760)   | $\ln(F)$                                        | 0.683***<br>(0.027)  | 1.445***<br>(0.419)  |
| $\Delta \text{Leverage}$            | 8.581*<br>(5.201)   | $1/\sigma_E$                                    | -1.763***<br>(0.266) | -0.974**<br>(0.480)  |
| $\Delta \text{PPENT/Total Asset}$   | -1.586<br>(10.840)  | $r_{it-1} - r_{mt-1}$                           | -1.339***<br>(0.192) | 0.556<br>(0.341)     |
| <i>CDS Active</i>                   | 2.432**<br>(1.069)  | <i>N/TA</i>                                     | -0.040***<br>(0.009) | -2.500<br>(9.361)    |
| Time fixed effects                  | yes                 | <i>CDS Firm</i>                                 | -2.210***<br>(0.722) |                      |
| Industry fixed effects              | yes                 | <i>CDS Active</i>                               | 2.378***<br>(0.736)  |                      |
| $R^2$                               | 9.75%               | <i>Number of Banks</i>                          | 0.153***<br>(0.035)  |                      |
| N                                   | 496                 | $\Delta \text{Number of Banks}$                 |                      | 0.043***<br>(0.016)  |
|                                     |                     | <i>CDS Active Odds Ratio</i>                    | 10.783               |                      |
|                                     |                     | <i>CDS Active Marginal Effect</i>               | 0.33%                |                      |
|                                     |                     | Time fixed effects                              | yes                  | yes                  |
|                                     |                     | Industry fixed effects                          | yes                  | yes                  |
|                                     |                     | Clustered standard error                        | yes                  |                      |
|                                     |                     | Pseudo $R^2$                                    | 24.32%               | 33.54%               |
|                                     |                     | N                                               | 658,966              | 480                  |
|                                     |                     | % of CDS observations                           | 10.83%               | 100%                 |
|                                     |                     | % of CDS bankruptcies                           | 4.47%                | 100%                 |
|                                     |                     | Sample probability of a bankruptcy              | 0.14%                | 5.83%                |

This table shows the results of an analysis of the impact of credit default swaps (CDS) on firm-creditor relationships. The creditor relationships are measured by bank relationships obtained from Dealscan LPC. For each firm, on a given date, we look back five years for any syndicated loan facilities extended to this firm. Summing over all such active facilities, we compute, on each date, the number of unique bank relationships. Panel A investigates the effect of CDS on the number of lenders to the firm. Panel B shows the effect of bank relationships on the bankruptcy risk of the firm. Model 1 in panel B conducts the analysis in a sample including firms with and without CDS traded. Model 2 in panel B conducts the analysis in the CDS sample.  $\Delta \text{Number of Banks}$  is the change in the number of bank relationships from one year before to two years after the inception of CDS trading.  $\Delta \ln(\text{Asset})$  is the change in the logarithm of the firm's total asset value.  $\Delta \text{ROA}$  is the change in the firm's return on assets.  $\Delta \text{Leverage}$  is the change in leverage.  $\Delta \text{PPENT/Total Asset}$  is the change in the ratio

**Figure 7:** Table 9 from the paper: CDS, bank relationships, and bankruptcy risk.

## 6 Short summary

- The paper studies whether CDS trading feeds back into the default risk of the underlying borrower.
- Using North American CDS transaction data from 1997–2009, it finds that rating downgrades and bankruptcy become more likely after CDS inception.
- The authors address endogeneity using lender-side instruments, propensity-score matching, and FIML, and the positive CDS effect survives all of them.
- Mechanism tests show that bankruptcy risk rises with CDS notional exposure, is especially strong for already distressed firms, is stronger for no-restructuring contracts, and is associated with an expansion in the lender base.
- These results are most naturally consistent with the empty-creditor problem and creditor-coordination failure.
- The paper does not conclude that CDS necessarily reduce welfare; it concludes that they can increase borrower bankruptcy risk even while offering hedging benefits elsewhere in the system.

## **7 Conclusion**

### **7.1 Core conclusion (what the paper establishes)**

The paper establishes that the introduction of CDS trading on a firm's debt is associated with a significant increase in the firm's credit risk, measured by both rating downgrades and bankruptcy. This result remains after serious attempts to deal with nonrandom CDS selection.

### **7.2 Economic intuition**

The intuition is that CDS weaken the link between a creditor's legal control rights and its economic exposure to default. Once some creditors are hedged, they may become tougher in renegotiation or less willing to coordinate around a workout. Distressed firms then end up more exposed to bankruptcy rather than restructuring.

### **7.3 Takeaway**

The paper's main message is that derivatives can reshape real corporate outcomes by changing incentives inside the contracting chain. In this case, a market designed to insure against default can, under some conditions, make default itself more likely.